Introduction to Soldering

IEEE@UCI

Attendance

Team Kits Updates

- All parts are ready!
- We need every team to sign up for a 3 hour soldering session
    - 11 AM onwards Sat & Sun for the next 2 weeks
- We'll help you all reflow solder your board

SECTION I

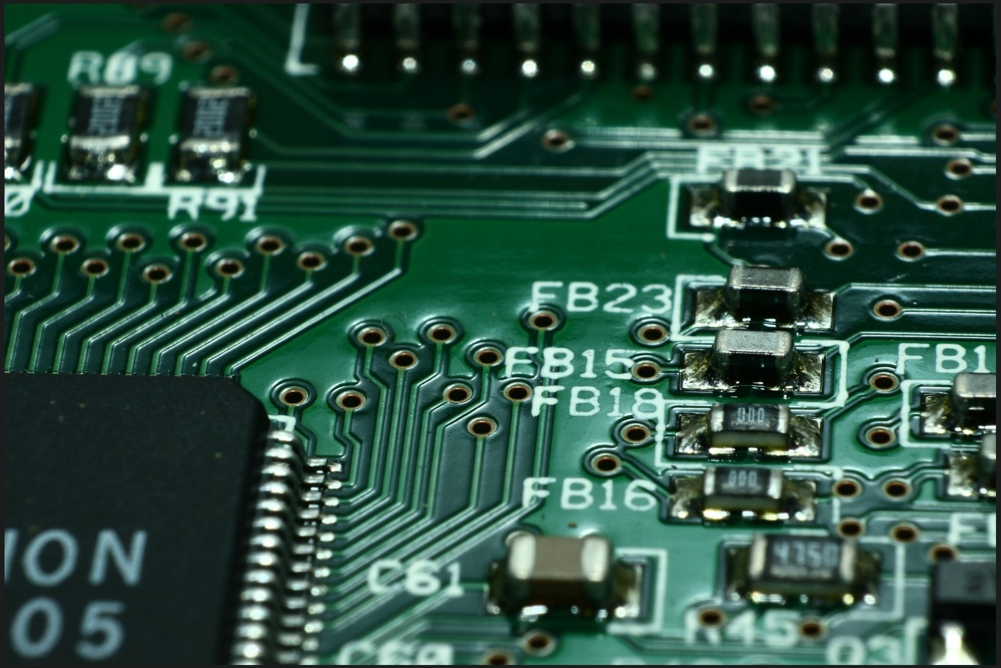
What is Soldering?

What is Soldering?

- Connects electrical joints
  - Wires, pads, pins, etc.
- Creates a reliable electrical connection
- Flexible process

Why is Soldering Important?

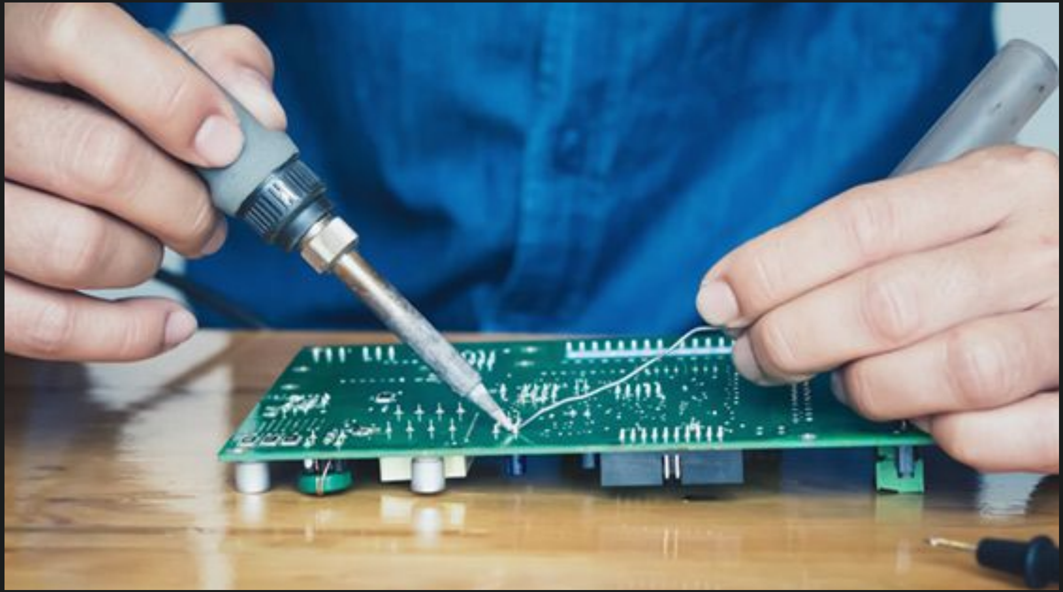
- Used in everyday electronics
- Quick, durable connections
- Components don’t have to be built into circuit boards



SECTION II

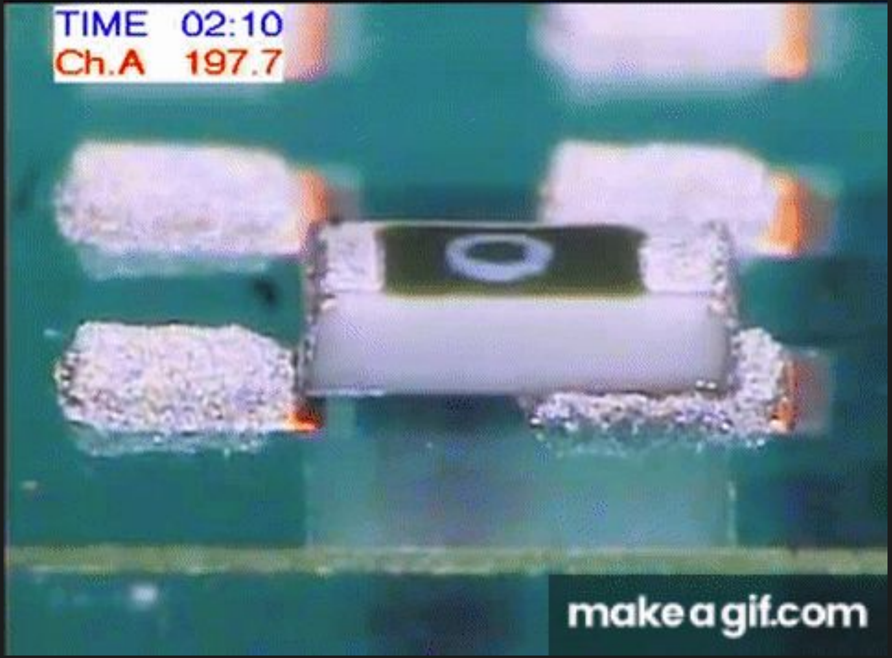
Types of Soldering

Hand Soldering



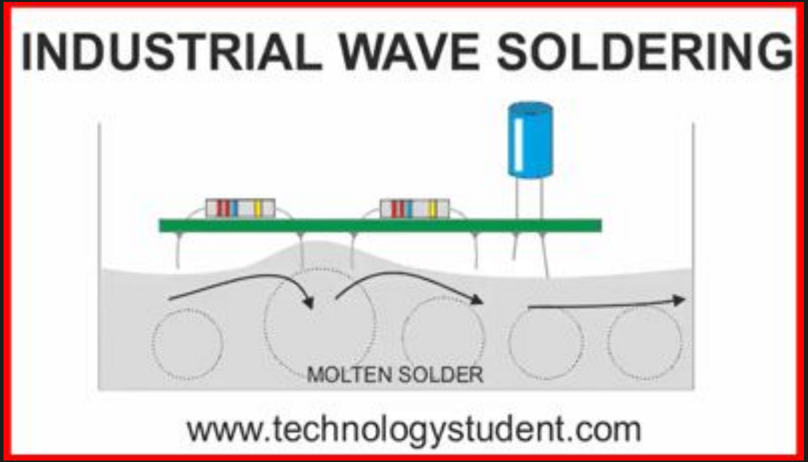
- Soldering iron heats up joints
- Solder wire melts
- Cools and joins joints
- Extremely flexible
- Great for 1-off jobs

Reflow Soldering



- Solder paste & flux applied to pads
- Components placed on pads
- Oven/hot plate/hot air melts solder
  - Snaps components into place
- Cools into joints
- Faster for large scale assembly
- Great for SMD

Wave Soldering

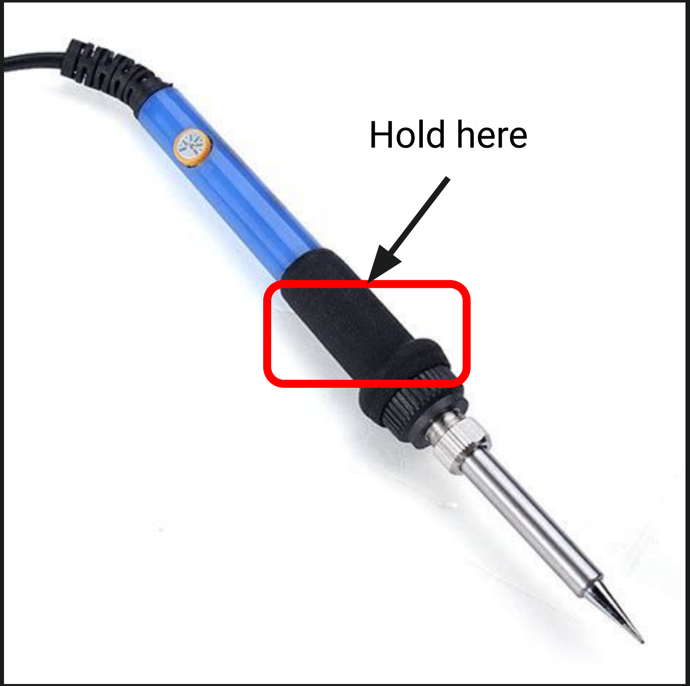


- Place through-hole components into their holes
- Run their pins over a wave of molten solder
- Solder clings onto pins and cools off to form joints
- Used in factories for through-hole components
- We will not be using this technique

SECTION III

Basics of Hand Soldering

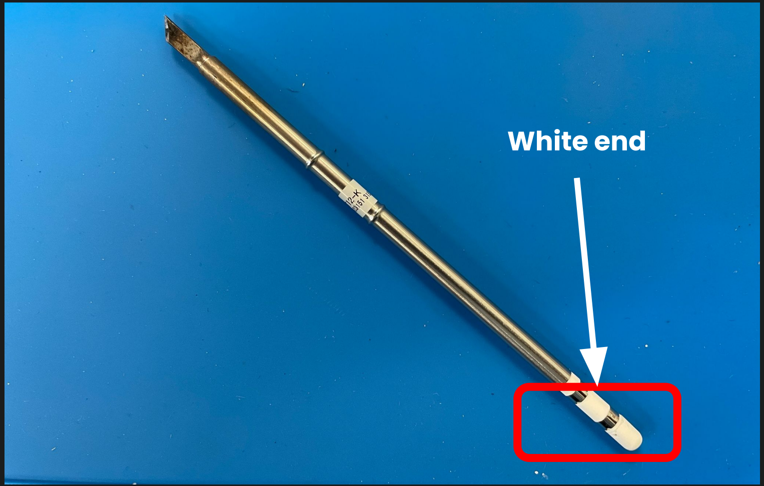
Soldering Iron



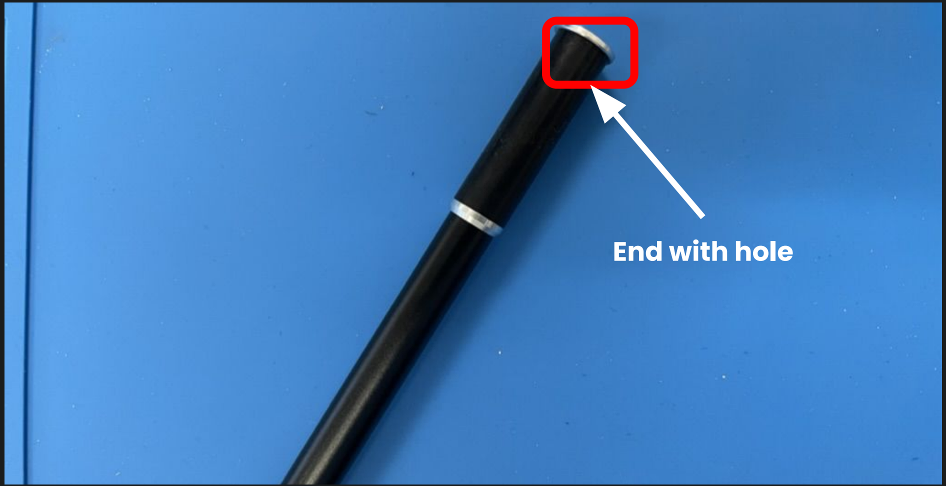
- Melts solder
- Hold like a pencil (on red rectangle)
- ***Do not hold the metal!***
- Make sure the tip doesn't touch anything while it is on

Soldering Iron (continued)

- This is the iron tip
- ***Do not touch when heated!***



- This is the iron handle
- The white end of the iron tip inserts into the hole of the iron body



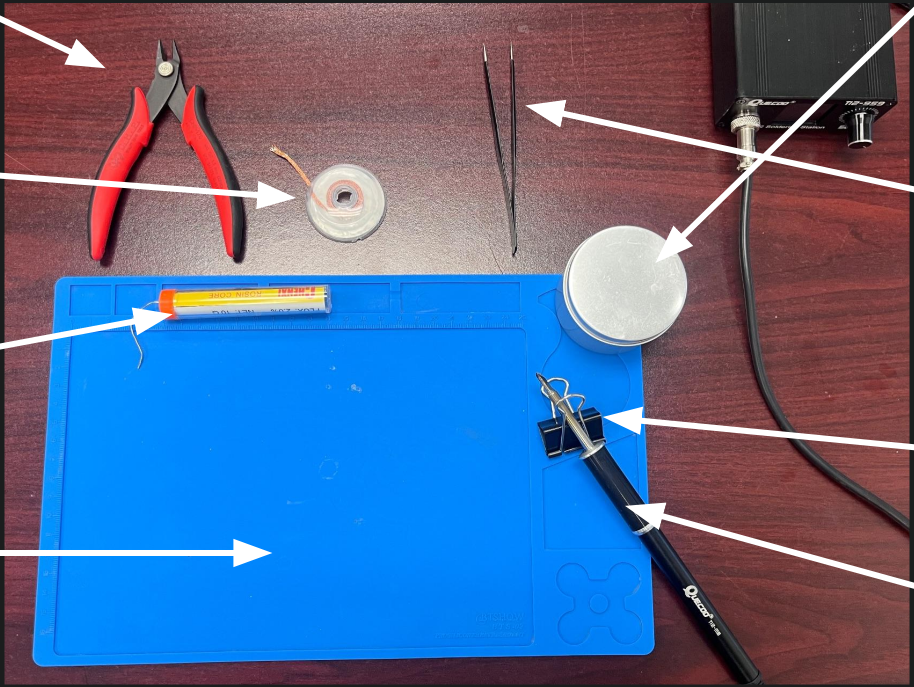
Station Layouts

Flush Cutters
- used to cut wires

Solder Wick
Cleans up excess solder and remove components

Solder Tube
- Dispenses the solder

Blue Silicone Mat -
your working area, keep all wire clippings and tools on it!



Brass Wool
- cleans oxidation and solder from tip of iron

Tweezers
Hold components and secure in place

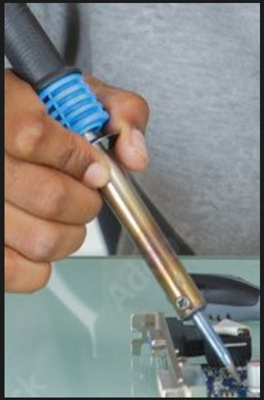
Binder Clip -
Holds the hot iron

Soldering Iron
- Heats the solder

Safety

- Never touch iron tip directly
- Always assume tip is hot
- Stow iron safely when not in use
- Don’t lick your hands (Flux and solder aren’t edible) (cancer alert)
- Components will be hot from soldering

Do not follow these images!

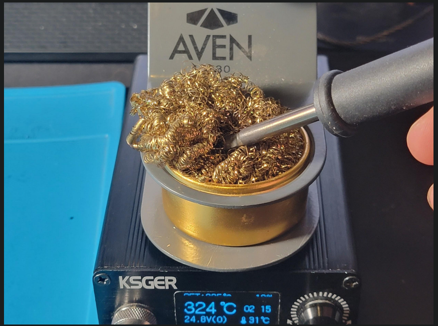


Safety Pt. 2

- Wear safety glasses (if not already wearing eye protection)
- Should be dressed in standard lab uniform (pants, shirt, closed toed shoes)
- For those with long hair, tie back (so you don’t lose it!)
- When dealing with resistors and components, feel free to bend the wires

How To - Tin Tip

Tin the Tip to prepare and protect from Oxidation



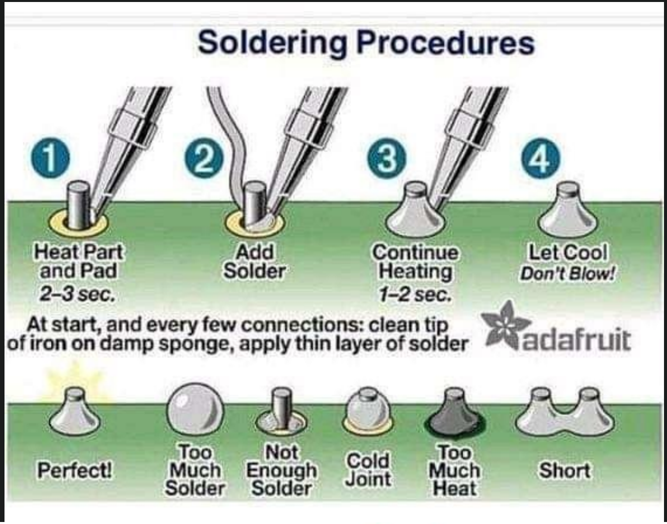
Steps

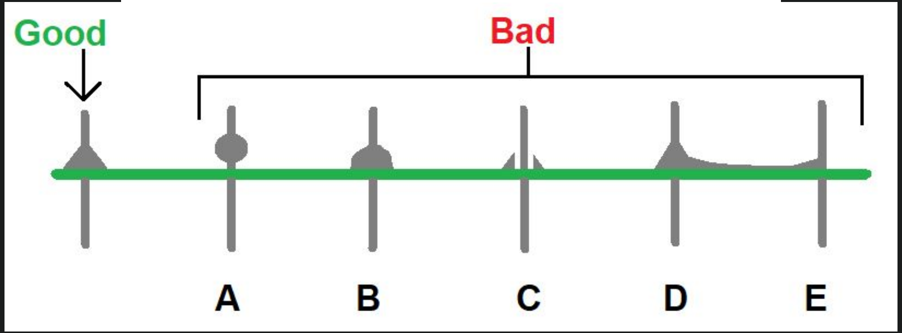
1. Prepare Board & Components
2. Assemble Components on Board
3. Confirm Circuit (Check with MM staff!)
4. Solder Components
5. Test Joints (with multimeter)
6. Cut Leads (extra wire)
7. Clean Up Area

- What's wrong with this image?

Basics of Soldering - Solder Joints

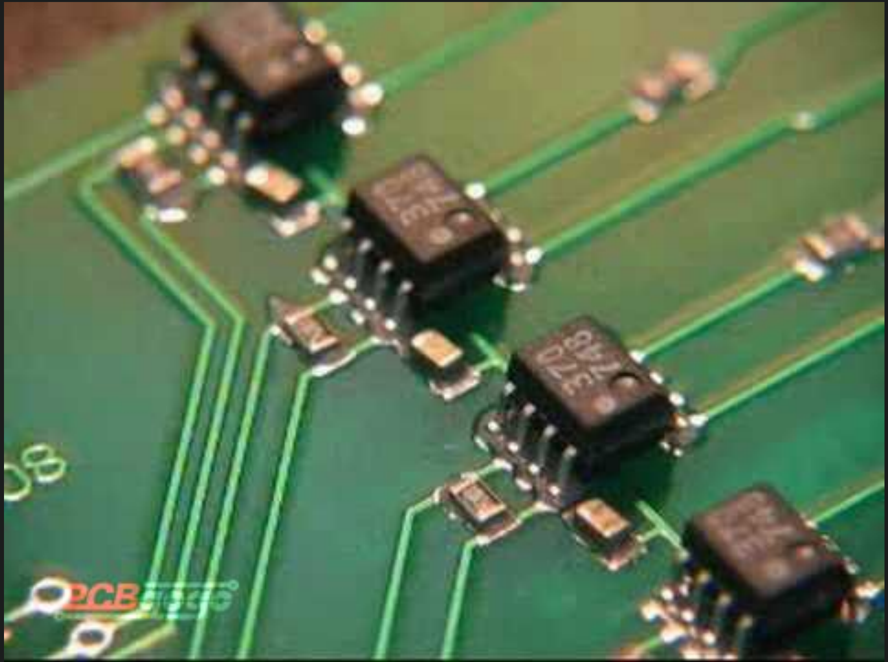
- We want OK solder joints
- Cold Joints = not enough heat to melt solder (increases resistance, could affect current)
- Insufficient Wetting = PCB/Components not clean



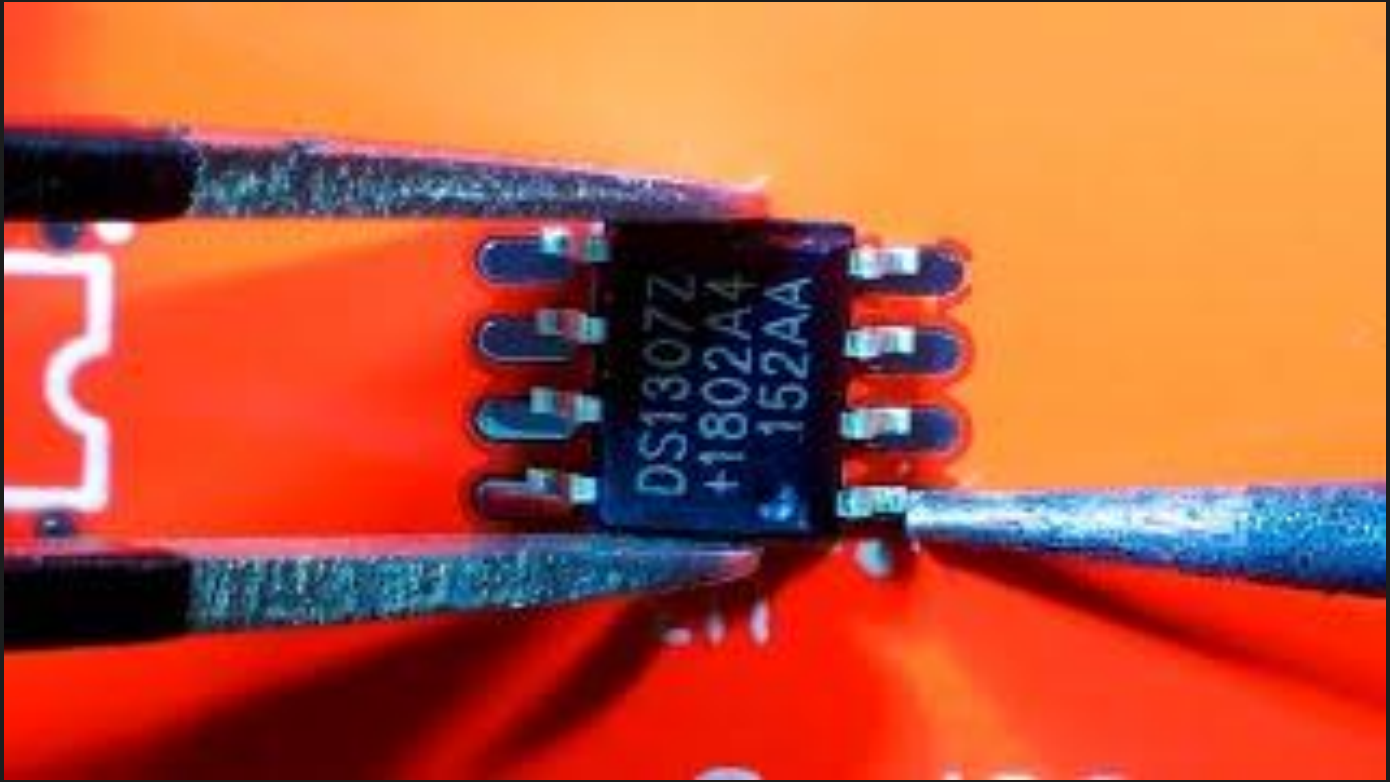


Tips & Tricks for Component Soldering

- Move from short to tall
- Resistors, Capacitors, Inductors first
  - Move to ICs, headers, regs after
- Do THT Soldering last
- Try not to melt plastic parts of components



SMD Soldering



Clean up:

Basics of Soldering - Soldering Correctly

Here is a short video that details the basics of good soldering!

SECTION IV

Basics of Reflow Soldering

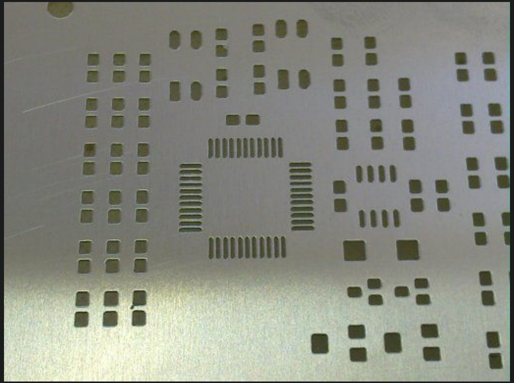
Equipment Necessary

Reflow Oven
* Melts solder and reflows PCB

Solder Paste
* Solder balls suspended in flux

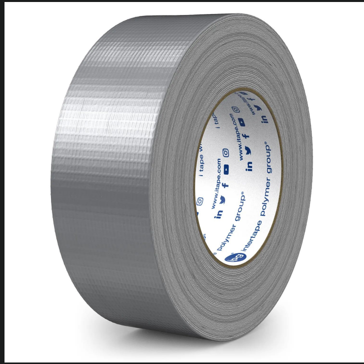
PCB Jig & Squeegee
* Holds PCB in place
* Spreads solder paste onto pads

Solder Stencil
* Mask
* Negative of PCB pad layout



Equipment Necessary (Continued)

Tweezers
* Allows for precise component placing



Tape
* Holds PCB & jig in place relative to stencil

Components
* Placed on PCB

Safety

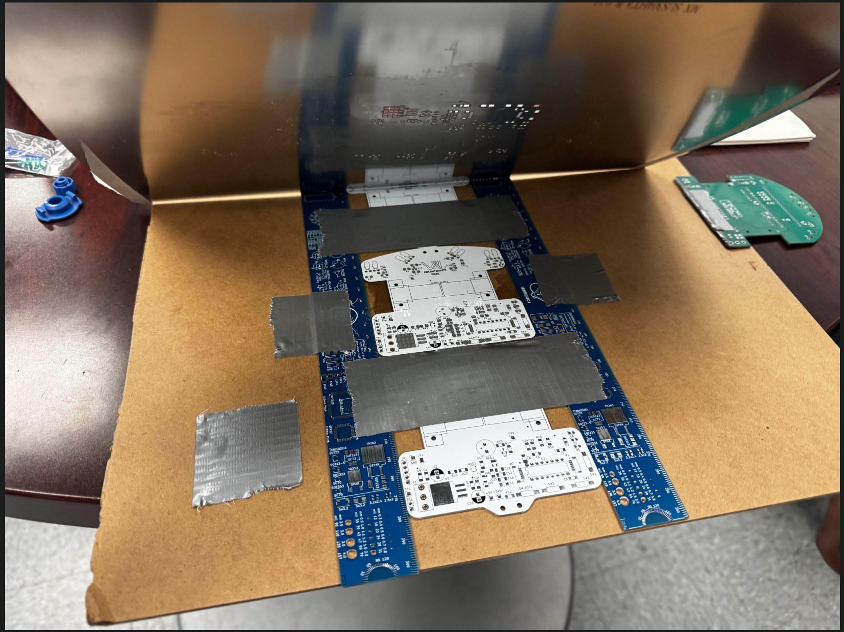
- Wash hands after touching solder
- Let PCB cool down after reflowing and before touching
- Be careful of stencil edge, it's VERY sharp
    - I've cut myself like 3 times :(it hurts

High Level Process

- Assemble jig around PCB and tape to hold shape
- Align stencil holes to PCB held in the jig
- Attach stencil w/ PCB in it to the jig
- Apply a line of solder paste on front of stencil
- Use squeegee to spread paste evenly over the PCB
- Remove PCB & place components onto the solder paste
- Place PCB with solder paste and components into the reflow oven and run it

Jig Example

- Rules & extra PCBs hold central PCB in place
- Central PCB is the final one
- All stencil components are taped together and to a board
- Stencil will also be restrained to jig

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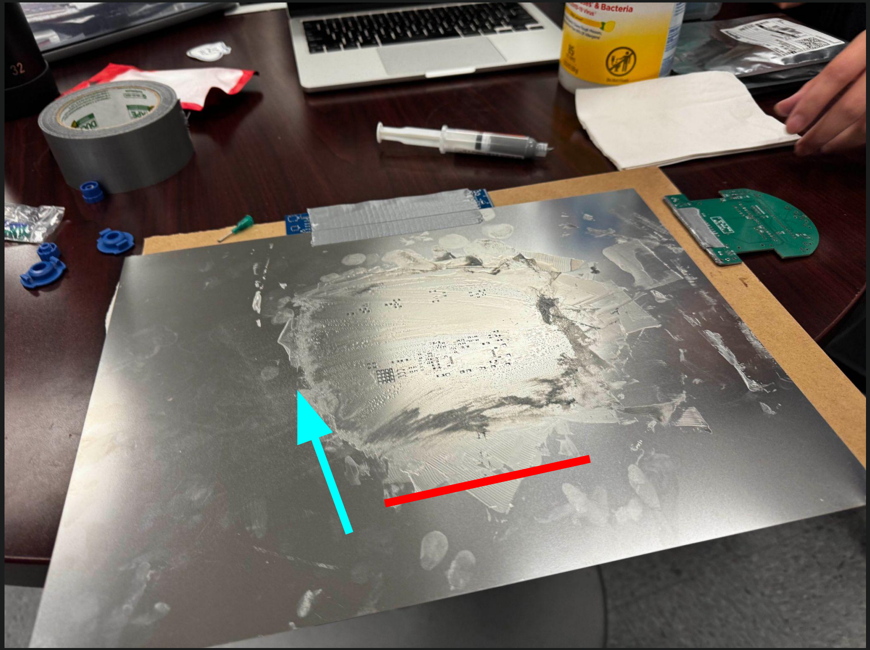
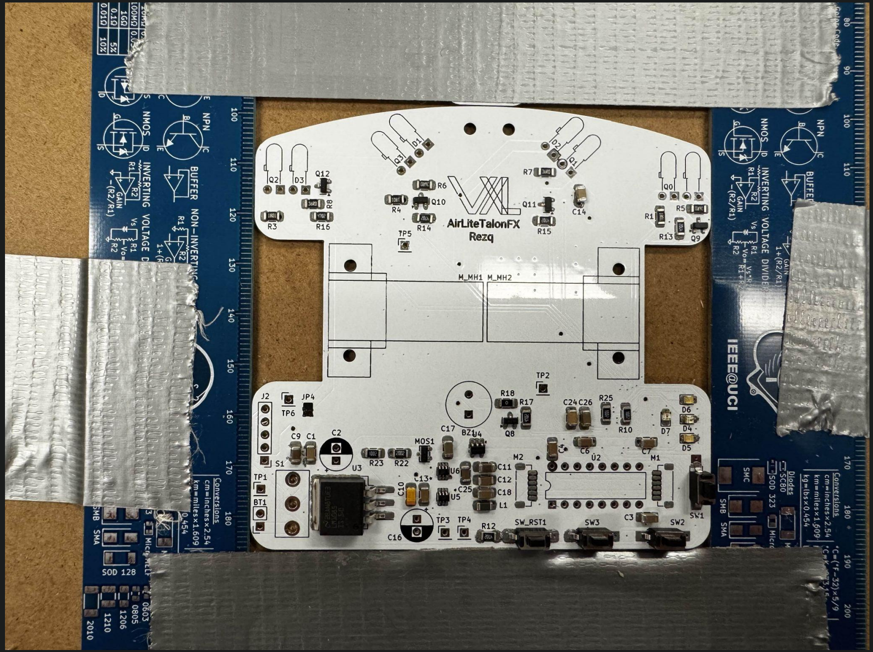
Paste Application Example

- Ensure stencil is PERFECTLY flat & in contact with the PCB
- Apply paste in a thin line (red)
- Squeegee over the PCB
- Redo process from a few different angles
- Ensure there is an even application
- Don’t want too much paste



Component Placement Example

- Remove stencil carefully
  - Ensure you don’t smear paste
- Place components 1-by-1 with tweezers



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